

Negative Parameters, Reciprocal Bases, and the Gaussian Boundary

New frontier results for the geometric-uniform, Fabius, inverse-Fabius, Rvachev, Thue–Morse, sinc-product, and Bell–Bernoulli framework

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Abstract

The current `ProveIt/Analysis/FabiusFunction/docs` corpus develops the Fabius and Rvachev functions from probability, functional equations, exact dyadic arithmetic, q -calculus, infinite sinc products, Thue–Morse cancellation, Legendre expansions, Lambert– W endpoint asymptotics, and formalized analysis. Its integrated frontier volume explicitly leaves open, for the geometric-uniform family, the density interpretation at negative parameters, centered affine transport, log-concavity and central plateaux, explicit Bernoulli cumulants and Bell moments in the generic parameter, and Gaussian-limit statements.

This report gives paper-level solutions or substantial advances for those gaps. For $0 < r < 1$, if

$$X_q = (1 - q) \sum_{n \geq 0} q^n U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1),$$

then the central identity is

$$X_{-r} - \frac{1}{2} \stackrel{d}{=} \frac{1+r}{1-r} \left(X_r - \frac{1}{2} \right).$$

It gives the exact negative-parameter CDF, quantile, support, smooth density, entropy scaling, sinc product, and all derivative transports. It also diagnoses a sign issue that was deliberately left uninterpreted in the repository: the previously selected total function at $q < 0$ is the *negative* of the Radon–Nikodym density. At $q = -1/2$ one obtains

$$F_{-1/2}(x) = F\left(\frac{x+1}{3}\right), \quad g_{-1/2}(x) = \frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right), \quad Q_{-1/2}(u) = 3F^{-1}(u) - 1,$$

which transports the complete Lambert– W_{-1} , Legendre, and Thue–Morse derivative structures to the support $[-1, 2]$.

A second exact duality concerns normalized finite geometric sums: reversal of the digits proves invariance under $q \mapsto q^{-1}$. Consequently the bases 2, -2 , 4, -4 , and all real $|q| > 1$, have canonical weak limits given by the reciprocal-base laws. A residue-class decomposition yields a multiplication formula for the laws and their infinite sinc products. The same framework proves log-concavity for every real $|q| < 1$, exact central plateaux when $|q| < 1/2$, and a Gaussian boundary as $|q| \uparrow 1$. The latter comes with an explicit Berry–Esseen bound, closed standardized Bernoulli cumulants, Bell-polynomial moment corrections, and a quartic characteristic-function expansion. Reproducible numerical experiments accompany the proofs.

All claims of novelty in this report mean *novel relative to the inspected repository corpus*; no claim of global publication priority is made.

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1 Scope, corpus audit, and novelty boundary

1.1 The inspected corpus

The review used the current integrated exposition `Fabius_Function_and_Rvachev_Up.tex`, the mathematical glossary, the non-elementarity article, the Lean walkthrough, the consolidated semi-formalized-research-frontiers.tex, the archive index and its historical families, and the TeX sources stored under papers. The archive is important for priority control: many early notes are frozen historical stages, while the two large integrated sources are the canonical current synthesis. The principal public entry points are [1, 2, 3, 4, 5].

The repository's own frontier box is unusually specific. It records that the law μ_q is absolutely continuous for every $|q| < 1$, but that the chosen function called f_q has only been identified with the Radon–Nikodym density for $0 < q < 1$. It separately lists centered affine transport, log-concavity, strictness, central plateaux, explicit Bernoulli cumulants, Bell moments, and Gaussian limits as unpromoted research layers [2]. These declarations make it possible to state a sharp non-duplication target.

1.2 What is taken as established

We use without reproving every foundational layer already organized in the repository:

- (i) existence and almost-sure convergence of the weighted uniform series;
- (ii) the affine independent-copy equation and uniqueness of its probability-law fixed point for every real $|q| < 1$, including negative q ;
- (iii) continuity and reflection symmetry of the CDF;
- (iv) exact support for the geometric law;
- (v) smoothness and the density formula for $0 < q < 1$;
- (vi) the half-base bridge $F_{1/2} = F$ and $f_{1/2}(x) = 2 \operatorname{up}(2x - 1)$;
- (vii) the Bernoulli/Bell, q -binomial, sinc-product, Legendre, and Lambert– W toolkits developed there.

The report nevertheless restates enough notation to make every new proof self-contained.

1.3 New results relative to the corpus

Frontier item	Result obtained here
Negative-parameter density	Exact affine duality with the positive parameter; explicit CDF, quantile, support and smooth density. The repository's selected total function is proved to equal minus the density when $q < 0$.
Centered transport	Complete coupling, characteristic-function, entropy, L^p , Sobolev, Legendre and endpoint-asymptotic transport.
Dyadic negative base	Exact formulas in terms of F , F^{-1} , and up , plus transferred Thue–Morse derivative peaks and Lambert– W_{-1} phase.
Shape	Log-concavity for all real $ q < 1$; exact central plateau for $0 \leq q < 1/2$, with negative-base interval and height in closed form.

Frontier item	Result obtained here
Multiplication structure	Residue-class decomposition into independent q^m -laws and the resulting multiplication law for cumulants and infinite sinc products.
Bases outside the unit interval	Exact finite $q \leftrightarrow q^{-1}$ reversal duality, canonical reciprocal-base limits, convergence bounds, and Bell/ q -bracket reciprocity.
Gaussian boundary	Standardized sign-invariance, a Berry–Esseen rate, all standardized Bernoulli cumulants, Bell moment corrections, and a quartic mod-Gaussian-type expansion.

2 Geometric-uniform preliminaries

2.1 The law, its centered form, and its transform

Let $(U_n)_{n \geq 0}$ be independent $\text{Unif}(0, 1)$ random variables. For real q with $|q| < 1$, define

$$X_q = (1 - q) \sum_{n=0}^{\infty} q^n U_n. \quad (1)$$

The series converges absolutely almost surely. Put

$$V_n = U_n - \frac{1}{2}, \quad Y_q = X_q - \frac{1}{2} = (1 - q) \sum_{n=0}^{\infty} q^n V_n. \quad (2)$$

The centering follows from $(1 - q) \sum_{n \geq 0} q^n = 1$. Every V_n is symmetric, which is the source of the negative-parameter duality.

Write

$$F_q(x) = \mathbb{P}(X_q \leq x), \quad \psi_q(t) = \mathbb{E} e^{itY_q}, \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Independence gives the centered infinite sinc product

$$\psi_q(t) = \prod_{n=0}^{\infty} \text{sinc} \left(\frac{(1 - q)q^n t}{2} \right). \quad (3)$$

The signs of the individual arguments disappear because sinc is even, but the factor $1 - q$ remains and will encode the affine stretch between $q = r$ and $q = -r$.

2.2 The affine fixed point

If X'_q is an independent copy of X_q , then splitting off the first digit yields

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q. \quad (4)$$

The repository proves that this equation uniquely characterizes the law among all probability measures for every $|q| < 1$, without a support or density assumption. In the positive range, X_r is supported on $[0, 1]$. At $r = 1/2$,

$$F_{1/2} = F, \quad g_{1/2}(x) = 2 \text{up}(2x - 1), \quad (5)$$

where F is the bounded Fabius function and up is the Rvachev up-function.

2.3 q -brackets, Bernoulli cumulants, and Bell polynomials

For an integer $m \geq 1$, set

$$[m]_q = 1 + q + \cdots + q^{m-1} = \frac{1 - q^m}{1 - q} \quad (6)$$

when $q \neq 1$, with the continuous value $[m]_1 = m$. We use the Bernoulli convention $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$. For a centered uniform $V \sim \text{Unif}(-1/2, 1/2)$,

$$\kappa_{2m}(V) = \frac{B_{2m}}{2m}, \quad \kappa_{2m+1}(V) = 0. \quad (7)$$

The complete exponential Bell polynomial $\mathcal{B}_n$ converts cumulants to moments:

$$\mathbb{E}Z^n = \mathcal{B}_n(\kappa_1(Z), \dots, \kappa_n(Z)). \quad (8)$$

3 Negative parameters: an exact affine duality

3.1 The coupling theorem

Theorem 3.1 (Negative/positive affine duality). *Let $0 < r < 1$ and*

$$c_r = \frac{1+r}{1-r}. \quad (9)$$

Then

$$\boxed{X_{-r} - \frac{1}{2} \stackrel{d}{=} c_r \left(X_r - \frac{1}{2} \right)}. \quad (10)$$

More strongly, there is a digitwise coupling under which the equality holds almost surely.

Proof. Let $V_n = U_n - 1/2$. Define reflected digits

$$V_n^- = (-1)^n V_n, \quad U_n^- = \frac{1}{2} + V_n^- = \begin{cases} U_n, & n \text{ even,} \\ 1 - U_n, & n \text{ odd.} \end{cases}$$

The sequence (U_n^-) is again i.i.d. uniform. Using it to construct the negative-base law gives, almost surely,

$$\begin{aligned} X_{-r} - \frac{1}{2} &= (1+r) \sum_{n \geq 0} (-r)^n V_n^- \\ &= (1+r) \sum_{n \geq 0} (-r)^n (-1)^n V_n \\ &= (1+r) \sum_{n \geq 0} r^n V_n \\ &= \frac{1+r}{1-r} \left((1-r) \sum_{n \geq 0} r^n V_n \right) \\ &= c_r \left(X_r - \frac{1}{2} \right). \end{aligned}$$

Absolute convergence justifies the termwise coupling. Forgetting the coupling proves the distributional identity. $\square$

Corollary 3.2 (Support, CDF, and quantile). *Let*

$$\ell_r = -\frac{r}{1-r}, \quad u_r = \frac{1}{1-r}. \quad (11)$$

Then $\text{supp}(X_{-r}) = [\ell_r, u_r]$, and for every real x ,

$$F_{-r}(x) = F_r\left(\frac{1}{2} + \frac{x - 1/2}{c_r}\right). \quad (12)$$

For generalized inverse quantiles,

$$Q_{-r}(u) = \frac{1}{2} + c_r \left(Q_r(u) - \frac{1}{2} \right), \quad 0 < u < 1. \quad (13)$$

Proof. The affine map $T_r(y) = 1/2 + c_r(y - 1/2)$ is increasing. It maps $[0, 1]$ onto $[\ell_r, u_r]$, and pushforward under an increasing affine map gives [equations \(12\) and \(13\)](#). $\square$

3.2 The correct negative-parameter density and the sign diagnosis

The integrated source defines, as a total symbolic function,

$$\tilde{f}_q(x) = \frac{F_q(x/q) - F_q((x - (1 - q))/q)}{1 - q}. \quad (14)$$

For $q > 0$ it proves that this is the density. At $q < 0$, division by q reverses the CDF interval. The next result resolves the open interpretation exactly.

Theorem 3.3 (Radon–Nikodym density for every nonzero parameter). *For $0 < |q| < 1$, the law of X_q has density*

$$g_q(x) = \frac{1}{1 - q} \mathbb{P}(x - (1 - q) \leq qX'_q \leq x), \quad (15)$$

where X'_q is an independent copy. Equivalently,

$$g_q(x) = \text{sgn}(q) \tilde{f}_q(x). \quad (16)$$

Thus for $-1 < q < 0$, the repository's selected total function is not a probability density but its negative. In particular it integrates to -1 , while $g_q \geq 0$ and integrates to 1.

Proof. Condition on $X'_q = y$ in [equation \(4\)](#). The conditional law is uniform on the interval $[qy, qy + 1 - q]$, of length $1 - q > 0$. Averaging the conditional density therefore gives [equation \(15\)](#). If $q > 0$, the event in that formula is

$$\frac{x - (1 - q)}{q} \leq X'_q \leq \frac{x}{q},$$

which gives $g_q = \tilde{f}_q$. If $q < 0$, the inequalities reverse and

$$g_q(x) = \frac{F_q((x - (1 - q))/q) - F_q(x/q)}{1 - q} = -\tilde{f}_q(x).$$

The CDF is continuous, so endpoint conventions do not matter. $\square$

Corollary 3.4 (Smooth negative density). *For $0 < r < 1$,*

$$g_{-r}(x) = \frac{1}{c_r} g_r\left(\frac{1}{2} + \frac{x - 1/2}{c_r}\right). \quad (17)$$

Consequently $g_{-r} \in C_c^\infty(\mathbb{R})$, with support $[\ell_r, u_r]$, and

$$g_{-r}^{(k)}(x) = c_r^{-(k+1)} g_r^{(k)}\left(\frac{1}{2} + \frac{x - 1/2}{c_r}\right). \quad (18)$$

Proof. Differentiate the affine CDF relation or use the change-of-variables formula for an affine pushforward. The derivative formula follows by the chain rule. $\square$

Corollary 3.5 (Norm and entropy transport). *For $1 \leq p \leq \infty$ and $k \geq 0$,*

$$\|g_{-r}^{(k)}\|_p = c_r^{1/p-k-1} \|g_r^{(k)}\|_p, \quad (19)$$

with $1/\infty = 0$. The differential entropy satisfies

$$h(X_{-r}) = h(X_r) + \log c_r. \quad (20)$$

The same affine law gives the standard Rényi-entropy scaling.

Proof. Use [equation \(18\)](#) and the substitution $y = 1/2 + (x - 1/2)/c_r$. The entropy identity is the usual Jacobian formula for $aX + b$ with $a = c_r > 0$. $\square$

3.3 Characteristic functions and sinc products

Corollary 3.6 (Centered transform scaling). *For all real t ,*

$$\psi_{-r}(t) = \psi_r(c_r t). \quad (21)$$

Equivalently,

$$\prod_{n \geq 0} \text{sinc}\left(\frac{(1+r)r^n t}{2}\right) = \prod_{n \geq 0} \text{sinc}\left(\frac{(1-r)r^n c_r t}{2}\right). \quad (22)$$

Proof. Apply characteristic functions to [equation \(10\)](#). The product identity is also immediate from $(1-r)c_r = 1+r$. $\square$

4 The exact negative dyadic law

4.1 Fabius, inverse Fabius, and Rvachev formulas

Set $r = 1/2$. Then $c_r = 3$, $\ell_r = -1$, and $u_r = 2$.

Theorem 4.1 (Negative dyadic bridge). *For every real x and every $0 < u < 1$,*

$$F_{-1/2}(x) = F\left(\frac{x+1}{3}\right), \quad (23)$$

$$g_{-1/2}(x) = \frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right), \quad (24)$$

$$Q_{-1/2}(u) = 3F^{-1}(u) - 1. \quad (25)$$

The support is $[-1, 2]$. The source-selected total function satisfies

$$\tilde{f}_{-1/2}(x) = -\frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right). \quad (26)$$

Proof. Insert $r = 1/2$ into [equations \(12\), \(13\) and \(17\)](#), then use the half-base bridge [equation \(5\)](#). The sign statement follows from [equation \(16\)](#). $\square$

This is a direct and unexpectedly economical link between a negative geometric base and the inverse Fabius function. It also means that every exact formula for F^{-1} or its asymptotics is automatically an exact formula for the negative dyadic quantile.

4.2 Thue–Morse derivative peaks on a stretched dyadic grid

The frontier corpus proves the sharp Rvachev derivative identities

$$\left\| \text{up}^{(m)} \right\|_{\infty} = 2^{\binom{m+1}{2}}, \quad \text{up}^{(m)}(c_{m,j}) = (-1)^{\text{wt}(j)} 2^{\binom{m+1}{2}}, \quad (27)$$

where

$$c_{m,j} = -1 + 2^{-m} + j2^{1-m}, \quad 0 \leq j < 2^m. \quad (28)$$

It also proves higher-order flatness at this grid: $\text{up}^{(k)}(c_{m,j}) = 0$ for $k > m$. Transport gives a new negative-base atlas.

Corollary 4.2 (Negative dyadic Thue–Morse atlas). *Define*

$$x_{m,j} = \frac{1 + 3c_{m,j}}{2} = -1 + \frac{3}{2^{m+1}} + \frac{3j}{2^m}. \quad (29)$$

Then

$$g_{-1/2}^{(m)}(x_{m,j}) = (-1)^{\text{wt}(j)} \left(\frac{2}{3}\right)^{m+1} 2^{\binom{m+1}{2}}, \quad (30)$$

$$\left\| g_{-1/2}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+1}{2} + m + 1}}{3^{m+1}}, \quad (31)$$

$$g_{-1/2}^{(k)}(x_{m,j}) = 0 \quad (k > m). \quad (32)$$

Thus the same Thue–Morse signs occur on a grid whose spacing is stretched from 2^{1-m} to $3 \cdot 2^{-m}$.

Proof. From [equation \(24\)](#),

$$g_{-1/2}^{(m)}(x) = \left(\frac{2}{3}\right)^{m+1} \text{up}^{(m)}\left(\frac{2x-1}{3}\right).$$

Set $(2x-1)/3 = c_{m,j}$ and use [equation \(27\)](#) and the higher-flatness result. $\square$

4.3 Legendre-series transport

Suppose the even Legendre expansion on $[-1, 1]$ is

$$\text{up}(t) = \sum_{n=0}^{\infty} b_n P_{2n}(t) \quad (33)$$

in L^2 , or in any stronger mode established for the coefficient sequence. Then the negative dyadic density has the immediate expansion

$$g_{-1/2}(x) = \frac{2}{3} \sum_{n=0}^{\infty} b_n P_{2n}\left(\frac{2x-1}{3}\right), \quad -1 \leq x \leq 2. \quad (34)$$

No coefficients need to be recomputed: only the Jacobian and affine Legendre coordinate change. More generally, if

$$g_r(y) = \sum_{n \geq 0} a_n^{(r)} P_{2n}(2y - 1)$$

on $[0, 1]$, then

$$g_{-r}(x) = \frac{1}{c_r} \sum_{n \geq 0} a_n^{(r)} P_{2n}\left(\frac{2x - 1}{c_r}\right). \quad (35)$$

This supplies a direct bridge from the positive- q Legendre theory to every negative parameter.

4.4 Exact transport of the Lambert- W_{-1} endpoint phase

Let $L = \log 2$, and let $\lambda(z)$ be the large positive solution of

$$z = \lambda 2^{-\lambda}. \quad (36)$$

For sufficiently small positive z ,

$$\lambda(z) = -\frac{1}{L} W_{-1}(-Lz). \quad (37)$$

The corpus emphasizes that periodic coefficient functions must be evaluated at this *exact* phase rather than at a phase with merely bounded error.

At the left endpoint of the negative dyadic support, write $x = -1 + \xi$. Then

$$F_{-1/2}(-1 + \xi) = F(\xi/3), \quad (38)$$

and the exact transported phase is

$$\boxed{\lambda_-(\xi) = \lambda(\xi/3) = -\frac{1}{L} W_{-1}\left(-\frac{L\xi}{3}\right).} \quad (39)$$

Therefore every all-orders Fabius endpoint expansion, including its smooth one-periodic logarithmic corrections, transfers verbatim after the replacement $\lambda \mapsto \lambda_-$ and the elementary Jacobian factors for derivatives. The right endpoint follows by reflection. For the inverse law, the exact relation is even simpler:

$$Q_{-1/2}(u) + 1 = 3F^{-1}(u). \quad (40)$$

5 Shape theory: log-concavity and exact plateaux

5.1 Log-concavity for all real parameters

Definition 5.1. A probability measure μ on $\mathbb{R}$ is log-concave if for nonempty compact sets A, B and $0 < \theta < 1$,

$$\mu(\theta A + (1 - \theta)B) \geq \mu(A)^\theta \mu(B)^{1-\theta}.$$

An integrable function is log-concave when its logarithm is concave on its positivity set, with the usual extended-value convention.

Theorem 5.2 (Log-concavity of the geometric-uniform family). *For every real q with $|q| < 1$, the law of X_q is log-concave. For $q \neq 0$, its smooth density g_q is log-concave on $\mathbb{R}$; at $q = 0$, the density is $\mathbf{1}_{[0,1]}$, which is log-concave in the extended sense.*

Proof. For finite N , the partial sum

$$X_q^{(N)} = (1 - q) \sum_{n=0}^N q^n U_n$$

is a convolution of uniform laws on compact intervals. A negative coefficient merely reflects an interval, so each factor remains log-concave. Affine images and convolutions preserve log-concavity by the Prékopa theorem [13]. Hence every finite law is log-concave. The partial sums converge almost surely, therefore weakly, to X_q . The class of log-concave probability measures is closed under weak convergence, so the limit law is log-concave. A log-concave measure that has a continuous density has that density as its log-concave representative. Smoothness for $q \neq 0$ follows from the positive theory and corollary 3.4; $q = 0$ is uniform. $\square$

Remark 5.3. This result supplies a global shape theorem that the repository lists as open. It implies unimodality and, together with reflection symmetry, places the mode at $1/2$. It does *not* by itself imply strict log-concavity; the next theorem exhibits a whole parameter region where strictness is impossible.

5.2 A dominant-digit plateau lemma

Lemma 5.4 (Uniform convolution plateau). *Let $U \sim \text{Unif}(0, 1)$, independent of a random variable Y supported on $[\alpha, \beta]$. Let $a > 0$ and $X = aU + Y$. If $a \geq \beta - \alpha$, then the density of X is exactly $1/a$ on*

$$[\beta, \alpha + a]. \quad (41)$$

When $a > \beta - \alpha$, this is a nondegenerate interval.

Proof. Conditioning on Y ,

$$f_X(x) = \frac{1}{a} \mathbb{P}(x - a \leq Y \leq x).$$

For $x \in [\beta, \alpha + a]$, one has $x \geq \beta$ and $x - a \leq \alpha$, so the event contains the whole support of Y and has probability one. $\square$

Theorem 5.5 (Exact positive and negative central plateaux). *Let $0 \leq r < 1/2$.*

For the positive law,

$$g_r(x) = \frac{1}{1 - r} \quad \text{for } r \leq x \leq 1 - r. \quad (42)$$

For the negative law,

$$g_{-r}(x) = \frac{1}{1 + r} \quad \text{for } \frac{r^2}{1 - r} \leq x \leq 1 - \frac{r^2}{1 - r}. \quad (43)$$

At $r = 1/2$ each interval collapses to the point $x = 1/2$. The plateau probability mass in either law is

$$\frac{1 - 2r}{1 - r}. \quad (44)$$

Proof. From the fixed-point equation,

$$X_r = (1 - r)U + rX'_r,$$

and rX'_r is supported on $[0, r]$. Apply lemma 5.4 with $a = 1 - r$, $\alpha = 0$, $\beta = r$. The condition $1 - r \geq r$ is exactly $r \leq 1/2$, and the height is $1/(1 - r)$.

For the negative law, apply the affine density transport [equation \(17\)](#) to the interval $[r, 1 - r]$. Its lower endpoint maps to

$$\frac{1}{2} + c_r \left(r - \frac{1}{2} \right) = \frac{r^2}{1 - r},$$

and symmetry gives the upper endpoint. The height is $c_r^{-1}(1 - r)^{-1} = 1/(1 + r)$. Multiplying height by interval length gives [equation \(44\)](#). $\square$

Conjecture 5.6 (Strict-log-concavity phase transition). *For every $1/2 \leq r < 1$, the density g_r is strictly log-concave on $(0, 1)$ in the multiplicative-inequality sense. For $0 < r < 1/2$, the interval $[r, 1 - r]$ is the maximal constant interval, and each shoulder is strictly log-concave. By affine duality, the corresponding statements hold for g_{-r} .*

The critical value $r = 1/2$ is structurally natural: it is exactly where the first uniform digit and the entire tail have equal support length. Proving strictness above the transition likely requires an equality-case analysis for repeated Prékopa convolution or a direct argument from the refinement equation.

6 Residue-class decomposition and multiplication formulas

6.1 Decomposition into independent q^m laws

Theorem 6.1 (Residue-class multiplication law). *Let $m \geq 1$ and $|q| < 1$. Let $X_{q^m}^{(0)}, \dots, X_{q^m}^{(m-1)}$ be independent copies of X_{q^m} . Then*

$$X_q \stackrel{d}{=} \frac{1}{[m]_q} \sum_{j=0}^{m-1} q^j X_{q^m}^{(j)}. \quad (45)$$

The centered variables satisfy the same weighted identity:

$$Y_q \stackrel{d}{=} \frac{1}{[m]_q} \sum_{j=0}^{m-1} q^j Y_{q^m}^{(j)}. \quad (46)$$

Proof. Split the defining series into residue classes $n = mk + j$:

$$\begin{aligned} X_q &= (1 - q) \sum_{j=0}^{m-1} q^j \sum_{k \geq 0} q^{mk} U_{mk+j} \\ &= \frac{1 - q}{1 - q^m} \sum_{j=0}^{m-1} q^j \left((1 - q^m) \sum_{k \geq 0} (q^m)^k U_{mk+j} \right). \end{aligned}$$

The parenthesized variables are independent copies of X_{q^m} , and $(1 - q)/(1 - q^m) = 1/[m]_q$. Since the weights sum to one, subtracting $1/2$ yields the centered identity. $\square$

Corollary 6.2 (Transform and cumulant multiplication). *For all real t ,*

$$\psi_q(t) = \prod_{j=0}^{m-1} \psi_{q^m} \left(\frac{q^j t}{[m]_q} \right). \quad (47)$$

Whenever the cumulant of order s is considered,

$$\kappa_s(X_q) = \frac{[m]_{q^s}}{[m]_q^s} \kappa_s(X_{q^m}). \quad (48)$$

Proof. Characteristic functions multiply over independent sums. Cumulants of order s scale by the s -th power of a coefficient and add, giving

$$\frac{1}{[m]_q^s} \sum_{j=0}^{m-1} q^{js} = \frac{[m]_{q^s}}{[m]_q^s}.$$

□

6.2 Even-odd splitting

For $m = 2$, [equation \(45\)](#) becomes

$$X_r \stackrel{\text{d}}{=} \frac{X_{r^2}^{(0)} + rX_{r^2}^{(1)}}{1+r}, \quad (49)$$

$$X_{-r} \stackrel{\text{d}}{=} \frac{X_{r^2}^{(0)} - rX_{r^2}^{(1)}}{1-r}. \quad (50)$$

Thus a negative-base law is a normalized difference of two independent positive-base laws. Combining this with [theorem 3.1](#) gives the distribution identity

$$\frac{X_{r^2}^{(0)} - rX_{r^2}^{(1)}}{1-r} - \frac{1}{2} \stackrel{\text{d}}{=} c_r \left(\frac{X_{r^2}^{(0)} + rX_{r^2}^{(1)}}{1+r} - \frac{1}{2} \right). \quad (51)$$

For centered transforms, ψ_{q^2} is even, so

$$\psi_{-r}(t) = \psi_{r^2}\left(\frac{t}{1-r}\right) \psi_{r^2}\left(\frac{rt}{1-r}\right). \quad (52)$$

Together with $\psi_{-r}(t) = \psi_r(ct)$, this yields the sinc-product identity

$$\boxed{\psi_r(ct) = \psi_{r^2}\left(\frac{t}{1-r}\right) \psi_{r^2}\left(\frac{rt}{1-r}\right)}. \quad (53)$$

At $r = 1/2$,

$$\boxed{\psi_{1/2}(3t) = \psi_{1/4}(2t)\psi_{1/4}(t)}. \quad (54)$$

This is precisely the interlacing of the even and odd factors of the dyadic infinite sinc product into two quarter-base products.

6.3 Computational significance

The residue formula provides a multiscale density algorithm. Choose m so that $|q|^m$ is small. The q^m -law then has a broad exact plateau and short refinement depth; convolving its independently scaled copies reconstructs g_q . This can outperform direct evaluation of a long sinc product when $|q|$ is close to one. For negative q , taking even m reduces every building block to a positive parameter.

7 Reciprocal bases and finite geometric normalization

7.1 Exact $q \leftrightarrow q^{-1}$ reversal

For an integer $N \geq 1$, define the normalized finite law

$$X_{q,N} = \frac{1}{[N]_q} \sum_{n=0}^{N-1} q^n U_n, \quad (55)$$

whenever $[N]_q \neq 0$. The coefficients sum to one, so $\mathbb{E}X_{q,N} = 1/2$. At $q = 1$, the continuous extension is the arithmetic mean $N^{-1} \sum U_n$.

Theorem 7.1 (Finite reciprocal-base duality). *Let $q \neq 0$ and suppose both normalizations are defined. Then*

$$\boxed{X_{q,N} \stackrel{d}{=} X_{1/q,N}}. \quad (56)$$

More strongly, if the digits on the reciprocal side are reversed, the two random variables are equal almost surely.

Proof. The q -bracket reciprocity identity is

$$[N]_{1/q} = q^{1-N} [N]_q. \quad (57)$$

Therefore

$$\frac{(1/q)^{N-1-n}}{[N]_{1/q}} = \frac{q^{-N+1+n}}{q^{1-N}[N]_q} = \frac{q^n}{[N]_q}.$$

The coefficient of U_n at base q is exactly the coefficient of the reversed digit U_{N-1-n} at base $1/q$. Since the digit vector is exchangeable, the laws agree; using the reversed vector gives pathwise equality. $\square$

7.2 Canonical limits for $|q| > 1$

Theorem 7.2 (Reciprocal-base compactification). *Let $|q| > 1$ and put $p = 1/q$. Then*

$$X_{q,N} \stackrel{d}{=} X_{p,N} \implies X_p \quad (N \rightarrow \infty). \quad (58)$$

Thus the normalized finite laws at every real base outside the unit interval have a canonical weak limit. Explicitly,

$$\begin{aligned} q = 2 : \quad & X_{q,N} \implies X_{1/2}, \\ q = -2 : \quad & X_{q,N} \implies X_{-1/2} \stackrel{d}{=} 3X_{1/2} - 1, \\ q = 4 : \quad & X_{q,N} \implies X_{1/4}, \\ q = -4 : \quad & X_{q,N} \implies X_{-1/4} \stackrel{d}{=} \frac{1}{2} + \frac{5}{3} \left(X_{1/4} - \frac{1}{2} \right). \end{aligned}$$

Proof. The reciprocal identity is [theorem 7.1](#). Since $|p| < 1$, $[N]_p \rightarrow (1-p)^{-1}$ and the normalized finite series converges almost surely to $(1-p) \sum_{n \geq 0} p^n U_n = X_p$. The special cases use [theorem 3.1](#). $\square$

Proposition 7.3 (Exponential coupling rate). *For $0 \leq p < 1$, there is a common-digit coupling such that*

$$|X_{p,N} - X_p| \leq p^N \quad \text{almost surely.} \quad (59)$$

For $p = -r$, $0 < r < 1$, and even N , there is a coupling such that

$$|X_{-r,N} - X_{-r}| \leq c_r r^N. \quad (60)$$

Hence the same bounds hold for the W_∞ Wasserstein distance.

Proof. For $p \geq 0$, write

$$A_N = (1-p) \sum_{n < N} p^n U_n, \quad T_N = (1-p) \sum_{n \geq N} p^n U_n.$$

Then $0 \leq A_N \leq 1 - p^N$, $0 \leq T_N \leq p^N$, $X_{p,N} = A_N/(1 - p^N)$, and $X_p = A_N + T_N$. Hence

$$X_{p,N} - X_p = \frac{p^N}{1 - p^N} A_N - T_N$$

is the difference of two numbers in $[0, p^N]$.

For negative $p = -r$, finite digit reflection gives

$$X_{-r,N} - \frac{1}{2} \stackrel{d}{=} c_{r,N} \left(X_{r,N} - \frac{1}{2} \right), \quad c_{r,N} = c_r \frac{1 - r^N}{1 - (-r)^N}.$$

When N is even, $c_{r,N} = c_r$. Couple both finite and infinite negative laws by the same affine images of the positive coupling, and apply [equation \(59\)](#). $\square$

7.3 Bell/ q -bracket reciprocity

Let $u_1 = 1/2$, and for $s \geq 2$ let u_s be the s -th cumulant of a uniform U , so $u_s = B_s/s$ for even s and $u_s = 0$ for odd $s > 1$.

Theorem 7.4 (Finite cumulants and moments). *For every admissible q, N ,*

$$\boxed{\kappa_s(X_{q,N}) = u_s \frac{[N]_{q^s}}{[N]_q^s}.} \quad (61)$$

Consequently

$$\mathbb{E}X_{q,N}^k = \mathcal{B}_k \left(u_1, u_2 \frac{[N]_{q^2}}{[N]_q^2}, \dots, u_k \frac{[N]_{q^k}}{[N]_q^k} \right). \quad (62)$$

Every ratio in [equation \(61\)](#) is invariant under $q \mapsto q^{-1}$, giving an algebraic Bell-polynomial proof of the moment part of [theorem 7.1](#).

Proof. Cumulants add over independent variables and scale homogeneously, so

$$\kappa_s(X_{q,N}) = u_s \sum_{n=0}^{N-1} \left(\frac{q^n}{[N]_q} \right)^s = u_s \frac{[N]_{q^s}}{[N]_q^s}.$$

Apply the moment–cumulant Bell formula. Finally,

$$[N]_{q^{-s}} = q^{s(1-N)} [N]_{q^s}$$

and $[N]_{q^{-1}}^s = q^{s(1-N)} [N]_{q^s}$, so the ratio is reciprocal-invariant. $\square$

This reciprocity is the probability-law analogue of the involution underlying Gaussian binomial reciprocity. It suggests a systematic dictionary between finite geometric uniform sums, symmetric functions of the coefficient multiset, and denominator-free q -identities.

8 The Gaussian boundary $|q| \uparrow 1$

8.1 Sign-invariant standardization

For $-1 < q < 1$,

$$\sigma_q^2 = \text{Var}(X_q) = \frac{(1-q)^2}{12(1-q^2)} = \frac{1-q}{12(1+q)}. \quad (63)$$

Define

$$Z_q = \frac{X_q - 1/2}{\sigma_q}. \quad (64)$$

Proposition 8.1 (Standardized sign collapse). *For $0 < r < 1$,*

$$\boxed{Z_{-r} \stackrel{d}{=} Z_r.} \quad (65)$$

Thus the standardized family depends only on $r = |q|$.

Proof. From [equation \(63\)](#), $\sigma_{-r} = c_r \sigma_r$. Divide [equation \(10\)](#) by this identity. $\square$

The centered characteristic function of Z_r has the useful normalized product

$$\Psi_r(t) = \mathbb{E} e^{itZ_r} = \prod_{n=0}^{\infty} \text{sinc}\left(\sqrt{3(1-r^2)} r^n t\right). \quad (66)$$

8.2 A quantitative central limit theorem

Theorem 8.2 (Berry–Esseen Gaussian boundary). *Let C_{BE} be any universal constant valid in the classical Berry–Esseen inequality for independent, non-identically distributed summands. Then for $0 < r < 1$,*

$$\boxed{\sup_{x \in \mathbb{R}} |\mathbb{P}(Z_r \leq x) - \Phi(x)| \leq C_{\text{BE}} \frac{3\sqrt{3}}{4} \frac{(1-r^2)^{3/2}}{1-r^3}.} \quad (67)$$

The same bound holds for Z_{-r} . In particular,

$$Z_{\pm r} \implies N(0, 1) \quad (r \uparrow 1), \quad (68)$$

and the right side of [equation \(67\)](#) is

$$C_{\text{BE}} \sqrt{\frac{3}{2}} \sqrt{1-r} (1 + \mathcal{O}(1-r)). \quad (69)$$

Proof. Write $Y_q = \sum_{n \geq 0} a_{n,q} V_n$ with $a_{n,q} = (1-q)q^n$ and $r = |q|$. Since

$$\mathbb{E} |V_n|^3 = \frac{1}{32}, \quad \text{Var}(V_n) = \frac{1}{12},$$

we have

$$\begin{aligned} \sum_{n \geq 0} |a_{n,q}|^3 &= \frac{|1-q|^3}{1-r^3}, \\ \sum_{n \geq 0} a_{n,q}^2 &= \frac{(1-q)^2}{1-r^2}. \end{aligned}$$

The Berry–Esseen Lyapunov ratio is therefore

$$\begin{aligned} \frac{\sum_n \mathbb{E} |a_{n,q} V_n|^3}{(\sum_n \text{Var}(a_{n,q} V_n))^{3/2}} &= \frac{(1/32)}{(1/12)^{3/2}} \frac{(1-r^2)^{3/2}}{1-r^3} \\ &= \frac{3\sqrt{3}}{4} \frac{(1-r^2)^{3/2}}{1-r^3}. \end{aligned}$$

Apply the finite-sum inequality and pass to the almost-sure limit. The sign collapse [equation \(65\)](#) handles $-r$. Expanding $1-r^2 \sim 2(1-r)$ and $1-r^3 \sim 3(1-r)$ gives [equation \(69\)](#). $\square$

8.3 Closed standardized Bernoulli cumulants

Theorem 8.3 (All standardized cumulants). *For $m \geq 1$, the even standardized cumulant of either Z_r or Z_{-r} is*

$$\lambda_{2m}(r) := \kappa_{2m}(Z_r) = \frac{B_{2m}}{2m} 12^m \frac{(1-r^2)^m}{1-r^{2m}}. \quad (70)$$

All odd cumulants vanish. In particular,

$$\lambda_4(r) = -\frac{6}{5} \frac{1-r^2}{1+r^2}, \quad (71)$$

$$\lambda_6(r) = \frac{48}{7} \frac{(1-r^2)^2}{1+r^2+r^4}. \quad (72)$$

For fixed $m \geq 2$, $\lambda_{2m}(r) = \mathcal{O}((1-r)^{m-1})$.

Proof. The cumulant of order $2m$ of $X_r - 1/2$ is

$$\frac{B_{2m}}{2m} (1-r)^{2m} \sum_{n \geq 0} r^{2mn} = \frac{B_{2m}}{2m} \frac{(1-r)^{2m}}{1-r^{2m}}.$$

Divide by

$$\sigma_r^{2m} = \left(\frac{1-r}{12(1+r)} \right)^m$$

to obtain [equation \(70\)](#). Odd cumulants vanish by symmetry. The negative law has the same standardized distribution. Factoring $1-r^{2m} = (1-r^2)(1+r^2+\dots+r^{2m-2})$ gives the order estimate. $\square$

8.4 Bell-polynomial moment correction

Corollary 8.4 (First correction to Gaussian moments). *For every fixed integer $m \geq 2$,*

$$\mathbb{E} Z_r^{2m} = (2m-1)!! + \binom{2m}{4} (2m-5)!! \lambda_4(r) + \mathcal{O}((1-r)^2) \quad (73)$$

$$= (2m-1)!! - \frac{6}{5} \binom{2m}{4} (2m-5)!! \frac{1-r^2}{1+r^2} + \mathcal{O}((1-r)^2). \quad (74)$$

All odd moments vanish.

Proof. Use the moment–cumulant Bell expansion. The Gaussian term consists entirely of pair blocks and equals $(2m-1)!!$. At first order in $1-r$, the only possible non-Gaussian partition has one block of size four and $m-2$ pairs. Its number is

$$\binom{2m}{4} (2m-5)!!.$$

Every block of size at least six contributes $\mathcal{O}((1-r)^2)$, as does a product of two fourth cumulants. Insert [equation \(71\)](#). $\square$

8.5 Quartic characteristic correction

Theorem 8.5 (First mod-Gaussian-type correction). *For every fixed $T > 0$, uniformly for $|t| \leq T$,*

$$\log \Psi_r(t) = -\frac{t^2}{2} - \frac{1-r^2}{20(1+r^2)}t^4 + \mathcal{O}_T((1-r)^2). \quad (75)$$

Consequently, for every fixed real t ,

$$\lim_{r \uparrow 1} \frac{\log \Psi_r(t) + t^2/2}{1-r} = -\frac{t^4}{20}. \quad (76)$$

Proof. Use [equation \(66\)](#) and the local expansion

$$\log \operatorname{sinc} z = -\frac{z^2}{6} - \frac{z^4}{180} + \mathcal{O}(z^6).$$

For $z_n = \sqrt{3(1-r^2)}r^nt$,

$$\begin{aligned} \sum_{n \geq 0} z_n^2 &= 3t^2, \\ \sum_{n \geq 0} z_n^4 &= 9t^4 \frac{(1-r^2)^2}{1-r^4} = 9t^4 \frac{1-r^2}{1+r^2}, \\ \sum_{n \geq 0} |z_n|^6 &= 27|t|^6 \frac{(1-r^2)^3}{1-r^6} = \mathcal{O}_T((1-r)^2). \end{aligned}$$

For r close enough to one, all z_n lie in a fixed neighborhood where the sixth-order remainder is uniform. Substitution gives [equation \(75\)](#). Finally, $(1-r^2)/(1+r^2) = (1-r) + \mathcal{O}((1-r)^2)$. $\square$

8.6 Expected Edgeworth and entropy refinements

Let H_k be the probabilists' Hermite polynomial. Fourier inversion of the formal cumulant expansion predicts

$$f_{Z_r}(z) = \varphi(z) \left[1 + \frac{\lambda_4(r)}{24} H_4(z) \right] + \mathcal{O}((1-r)^2), \quad (77)$$

and

$$\mathbb{P}(Z_r \leq z) = \Phi(z) - \frac{\lambda_4(r)}{24} H_3(z) \varphi(z) + \mathcal{O}((1-r)^2). \quad (78)$$

The fixed-frequency theorem above is rigorous; making the remainders uniform in z requires a high-frequency estimate uniform in r , which is not supplied here.

Conjecture 8.6 (Uniform first Edgeworth expansion). *The remainders in [equations \(77\)](#) and [\(78\)](#) are $\mathcal{O}((1-r)^2)$ uniformly in z , with the density statement remaining valid after any fixed number of derivatives when multiplied by a suitable polynomial weight. The second-order term is the standard Bell–Hermite combination of $\lambda_6 H_6/720$ and $\lambda_4^2 H_8/(2 \cdot 24^2)$.*

Conjecture 8.7 (Entropy deficit). *Let $D(\mathcal{L}(Z_r) \| N(0, 1))$ be relative entropy. Then*

$$D(\mathcal{L}(Z_r) \| N(0, 1)) = \frac{\lambda_4(r)^2}{48} + o((1-r)^2) = \frac{3}{100}(1-r)^2 + o((1-r)^2). \quad (79)$$

Equivalently,

$$h(Z_r) = \frac{1}{2} \log(2\pi e) - \frac{3}{100}(1-r)^2 + o((1-r)^2).$$

The coefficient $1/48$ follows formally by inserting $f/\varphi = 1 + (\lambda_4/24)H_4 + \dots$ into relative entropy and using $\mathbb{E}_\varphi H_4^2 = 4!$. A proof should combine the uniform Edgeworth expansion with tail control supplied by log-concavity.

9 Numerical experiments

9.1 Reproducibility and deterministic error control

The accompanying script `numerical_experiments.py` uses NumPy, SciPy, and Matplotlib with seed 20260830. It truncates [equation \(1\)](#) only after the deterministic tail bound

$$\left| (1-q) \sum_{n \geq N} q^n U_n \right| \leq \frac{|1-q| |q|^N}{1-|q|} \quad (80)$$

falls below the requested tolerance. The coupling checks are stronger than Monte Carlo comparisons: reflected or reversed digits make the asserted equality exact, so the reported errors measure floating-point roundoff.

Table 2: Digit-reflection coupling for the negative/positive affine duality.

r	samples	terms	maximum coupling error
0.20	40000	21	7.772e-16
0.50	40000	46	1.776e-15
0.80	40000	137	6.661e-15

Table 3: Reversal coupling for finite reciprocal-base duality.

q	N	samples	maximum coupling error
2	14	35000	4.441e-16
-2	14	35000	8.882e-16
4	14	35000	4.441e-16
-4	14	35000	8.882e-16

Table 4: Standardized geometric-uniform laws approaching the Gaussian boundary.

r	samples	KS distance	empirical excess	exact excess
0.50	260000	0.0282	-0.71908	-0.72000
0.80	230000	0.0080	-0.26110	-0.26341
0.95	180000	0.0022	-0.05054	-0.06150
0.98	120000	0.0016	-0.03496	-0.02424

9.2 Negative dyadic collapse

9.3 The plateau phase

9.4 Approach to the Gaussian

The empirical excess kurtosis becomes noisy very near $r = 1$, because its true value is small and fourth moments converge more slowly than CDFs. The exact values in [table 4](#) are analytic, not fitted.

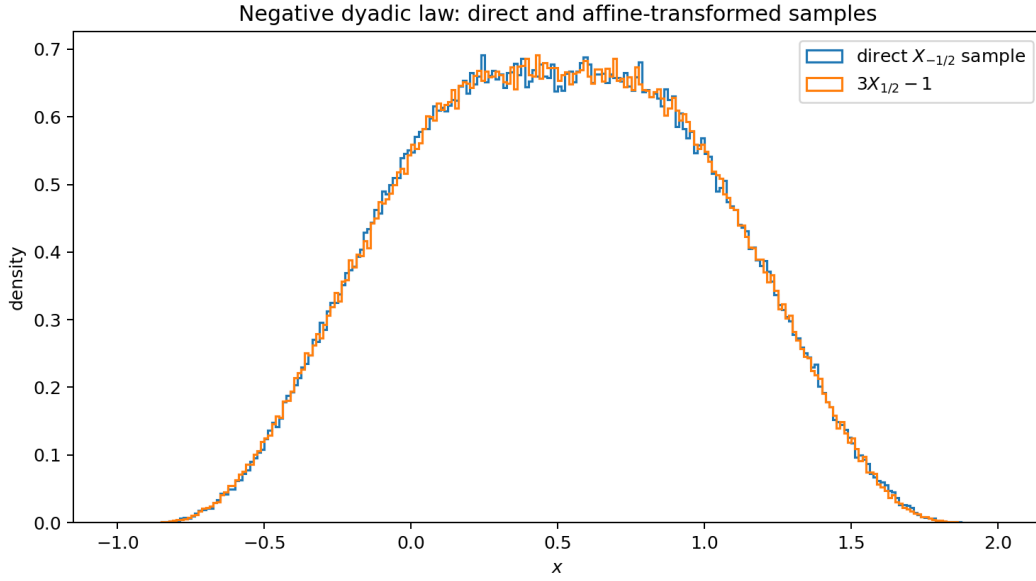


Figure 1: Independent simulations of $X_{-1/2}$ and $3X_{1/2} - 1$. The visual agreement supplements the exact digit-reflection coupling in [table 2](#).

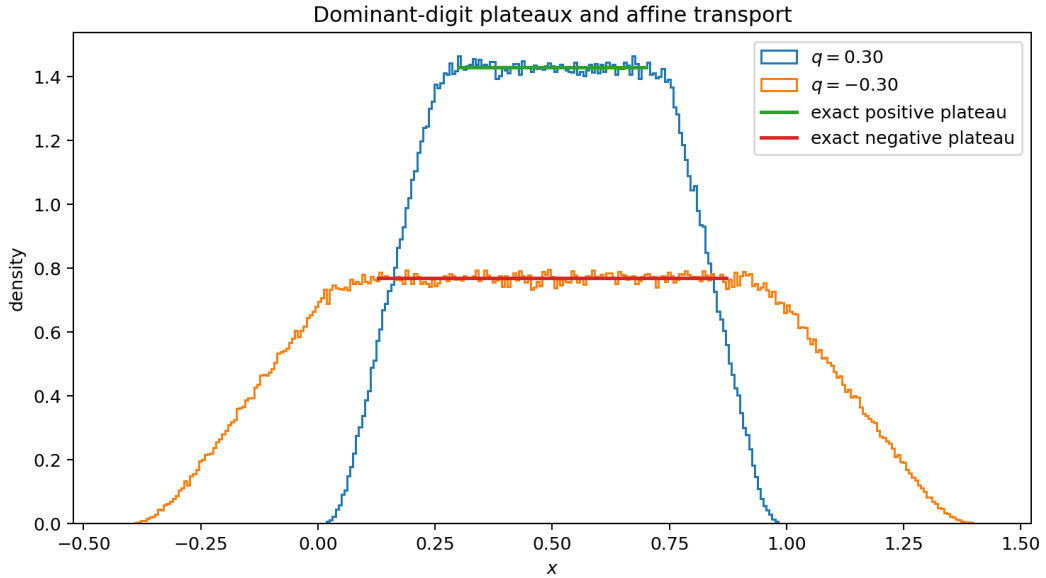


Figure 2: For $r = 0.30$, the exact positive plateau is $[0.30, 0.70]$ at height $1/0.70$, while the negative plateau is $[0.09/0.70, 1 - 0.09/0.70]$ at height $1/1.30$.

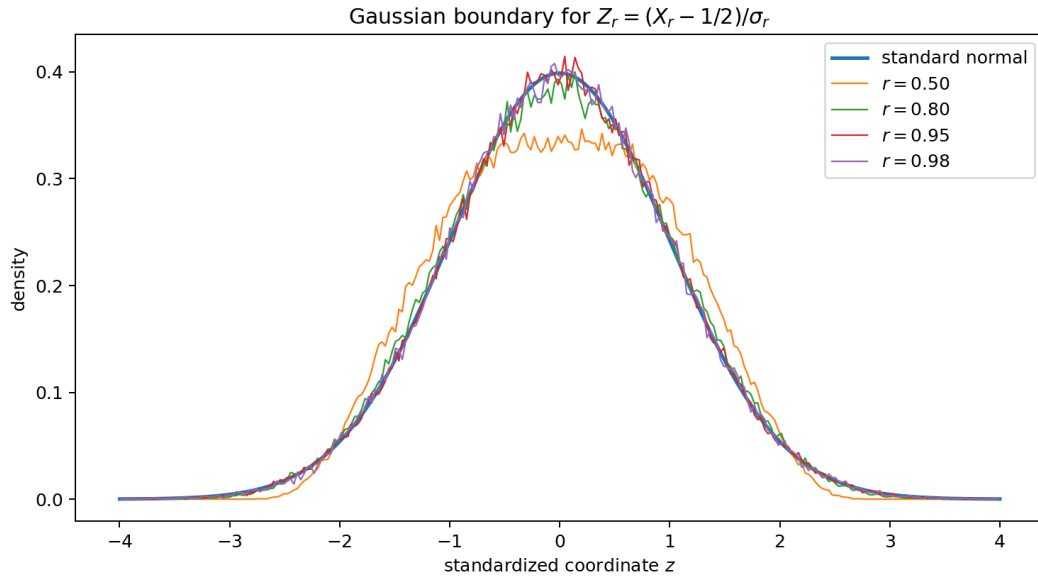


Figure 3: Standardized densities for increasing r , compared with the standard normal. The exact excess kurtosis is $-\frac{6}{5}(1-r^2)/(1+r^2)$, and the measured Kolmogorov distance decreases consistently with [theorem 8.2](#).

10 Further conjectures and frontier directions

10.1 All-orders Gaussian asymptotics

Conjecture 10.1 (Complete Bernoulli–Bell–Hermite expansion). *For every integer $M \geq 1$, the standardized density has an expansion through order $(1-r)^M$ whose coefficients are finite Bell polynomials in the exact cumulants $\lambda_4(r), \lambda_6(r), \dots, \lambda_{2M+2}(r)$, expressed in the Hermite basis. The expansion is uniform with polynomial weights and differentiable termwise a fixed number of times.*

A proof would turn the corpus’s Bernoulli and Bell machinery into a full parameter-edge asymptotic calculus. The crucial analytic estimate is uniform control of [equation \(66\)](#) outside the low-frequency window.

10.2 A generic- q Lambert- W endpoint theory

For fixed $0 < r < 1$, the negative Laplace transform of X_r is

$$\mathbb{E}e^{-sX_r} = \prod_{n \geq 0} \frac{1 - e^{-(1-r)r^n s}}{(1-r)r^n s}. \quad (81)$$

Its logarithm is a geometric harmonic sum. Mellin analysis should produce a quadratic polynomial in $\log s$, a smooth periodic fluctuation in $\log_{1/r} s$, and descending corrections. Saddle inversion then produces a Lambert- W_{-1} phase, just as in the dyadic theory.

Conjecture 10.2 (Generic geometric endpoint expansion). *For every fixed $0 < r < 1$, $F_r(x)$ admits an all-orders small- x saddle expansion whose coefficient functions are smooth and one-periodic in the exact logarithmic saddle phase. The leading inversion is expressible using W_{-1} , while the periodic phase must be kept exact. Under [theorem 3.1](#), the entire expansion transfers to F_{-r} at its left endpoint by $x \mapsto (x - \ell_r)/c_r$.*

This project would unify the corpus’s dyadic Lambert analysis with the generic geometric-uniform family and explain how the log-periodic oscillation changes with the base.

10.3 Legendre coefficients and endpoint saddles

Because each g_q is compactly supported and smooth, its affine Legendre coefficients decay faster than every power. The precise stretched-exponential envelope should be controlled by the same complex saddle geometry that governs endpoint flatness.

Conjecture 10.3 (Legendre–Lambert asymptotic bridge). *For the even Legendre coefficients of g_r , the logarithm of the coefficient magnitude has a complete asymptotic expansion whose principal scale is obtained by balancing the large-degree Legendre boundary layer with the Lambert saddle of F_r . A smooth log-periodic factor survives in the coefficient asymptotics. Negative-base coefficients obey exactly the same law after the affine normalization in [equation \(35\)](#).*

This would connect three parts of the existing program that are currently largely parallel: endpoint Lambert asymptotics, Legendre expansions, and infinite sinc products.

10.4 Reciprocal-base algebra and q -identities

The coefficient multiset proof of [theorem 7.1](#) suggests a broader principle: every symmetric polynomial or entire symmetric function of $\{q^n/[N]_q : 0 \leq n < N\}$ is reciprocal-invariant. Power sums give cumulants; elementary symmetric functions give Gaussian-binomial expressions.

Problem 10.4 (A probability proof of a full Gaussian-binomial reciprocity package). *Construct natural random variables or random polynomials whose elementary-symmetric statistics of the finite geometric weights recover $\begin{bmatrix} N \\ k \end{bmatrix}_q$, and derive $\begin{bmatrix} N \\ k \end{bmatrix}_{1/q} = q^{-k(N-k)} \begin{bmatrix} N \\ k \end{bmatrix}_q$ directly from digit reversal. Then seek mixed Bell/Gaussian identities obtained by converting between power sums and elementary symmetric functions.*

10.5 A formalization roadmap

The strongest immediate formal targets are elementary enough to be realistic additions to the existing Lean development:

1. define the centered geometric-uniform law and prove the reflected-digit coupling;
2. derive the affine pushforward equality $\mu_{-r} = T_{r*}\mu_r$;
3. prove the CDF, quantile, support, and smooth-density transports;
4. replace the negative-parameter density ambiguity by the sign-corrected theorem $g_q = \text{sgn}(q)\tilde{f}_q$;
5. formalize the residue-class multiplication law and its characteristic-function corollary;
6. prove finite reciprocal-base duality by coefficient extensionality and list $q = \pm 2, \pm 4$ wrappers;
7. formalize the exact plateau theorem from the CDF tails;
8. package Bernoulli cumulants and standardized sign collapse.

Log-concavity may require additional convex-measure infrastructure. Berry–Esseen and uniform Edgeworth theory are longer-term targets, but the exact cumulant and quartic formal-power-series identities can be separated from those analytic theorems.

11 Conclusions

The geometric-uniform family has more symmetry than is visible from the positive parameter alone. Centering turns every alternating geometric digit into a reflected uniform digit, and this collapses the entire negative half of the parameter interval to an affine copy of the positive half. That one observation resolves the negative density frontier, identifies the sign of the previously selected total function, and transports Fabius, inverse Fabius, Rvachev, Legendre, Lambert– W , sinc-product, and Thue–Morse structures in one step.

Two further reorganizations complete the parameter picture. Residue classes produce a q -multiplication law, while finite digit reversal identifies q with q^{-1} . Thus negative parameters, bases outside the unit interval, and smaller positive bases are not separate theories but affine or reciprocal views of the same laws.

Finally, the parameter edge $|q| \uparrow 1$ is a genuine Gaussian boundary. The rate is controlled by a closed Berry–Esseen ratio, and the departure from normality is encoded by Bernoulli cumulants and Bell partitions. The quartic coefficient $-t^4/20$ is the first term of an all-orders asymptotic program that should connect the probability family to the repository’s existing Bell, Bernoulli, Lambert, Legendre, and formal saddle machinery.

A Auxiliary algebra

A.1 Support endpoints under negative affine transport

Since $T_r(y) = 1/2 + c_r(y - 1/2)$,

$$T_r(0) = \frac{1 - c_r}{2} = -\frac{r}{1 - r}, \quad T_r(1) = \frac{1 + c_r}{2} = \frac{1}{1 - r}.$$

For the positive plateau endpoint $y = r$,

$$T_r(r) = \frac{1}{2} + \frac{1 + r}{1 - r} \left(r - \frac{1}{2} \right) = \frac{r^2}{1 - r}.$$

These identities explain the apparently asymmetric support and the symmetric central plateau of the negative law.

A.2 Finite negative/positive scale

Digit reflection in [equation \(55\)](#) gives

$$X_{-r,N} - \frac{1}{2} \stackrel{d}{=} \frac{[N]_r}{[N]_{-r}} \left(X_{r,N} - \frac{1}{2} \right),$$

where

$$\frac{[N]_r}{[N]_{-r}} = \frac{1 + r}{1 - r} \frac{1 - r^N}{1 - (-r)^N}.$$

For even N , this is exactly c_r ; for odd N , it converges exponentially to c_r .

A.3 Low standardized cumulants

The first values from [equation \(70\)](#) are

$$\begin{aligned} \lambda_2(r) &= 1, \\ \lambda_4(r) &= -\frac{6}{5} \frac{1 - r^2}{1 + r^2}, \\ \lambda_6(r) &= \frac{48}{7} \frac{(1 - r^2)^2}{1 + r^2 + r^4}, \\ \lambda_8(r) &= -\frac{432}{5} \frac{(1 - r^2)^3}{1 + r^2 + r^4 + r^6}. \end{aligned}$$

The last coefficient uses $B_8 = -1/30$. These feed directly into complete Bell polynomials and higher Edgeworth terms.

B Numerical file manifest

The archive accompanying this report contains:

- `fabius_frontier_report.tex` – this source;
- `fabius_frontier_report.pdf` – compiled report;
- `numerical_experiments.py` – fully commented reproducible code;
- `numerical_results.tex` – generated tables;

- three CSV files containing the machine-readable results;
- PDF and PNG versions of all figures;
- README.txt with compilation and execution commands.

References

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